OPERATING SYSTEMS – ASSIGNMENT 1

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TASK 11 – Recommend and Defend the Operating-System Plan (4 Marks)

Recommend the complete OS plan for the hospital, including type and structure for each environment, services, protection and I/O methods, limitations and risks, and suitability for three hospitals.

The recommendation must be based on analysis and measurements.

Alternative Concept and Complexity Analysis: If the whole system moved to VMs on two large servers, evaluate monitor response time, memory/processor overhead and reliability.

Evidence Expected: Technically justified plan supported by analysis and measurements.

ANSWER

11.1 Complete recommended architecture

11.2 Limitations and risks

RTOS selection alone does not prove the 10 ms requirement. End-to-end worst-case response must be tested on the actual monitor hardware with maximum interrupt load and CPU load.

Cluster fail-over may cause brief interruption, lost in-flight sessions or split-brain if quorum and fencing are incorrectly configured. Test fail-over and restore procedures.

Multiple OS families increase patching, training and support complexity. Use central asset inventory, configuration management, signed updates and controlled maintenance windows.

Network and cloud outages can prevent remote record access. Keep local alarm logic independent, cache only necessary data securely and define downtime procedures.

Legacy clinical devices may have unsupported drivers or software. Isolate them on segmented networks and plan replacement.

Security controls must include backup recovery, vulnerability management, monitoring, access reviews and incident response; encryption alone is insufficient.

11.3 Suitability for three hospitals

The design scales to three hospitals by using a distributed record-sharing layer with secure APIs, identity federation, common patient identifiers and agreed data formats. Each hospital should retain local services and a local operational mode so a WAN outage does not disable bedside alarms or all clinical work. Database replication must be designed around consistency requirements: strongly consistent transactions for critical updates where needed, and carefully governed asynchronous replication for disaster recovery or less time-sensitive sharing. Central governance should define access policies, audit retention, backup and recovery objectives.

11.4 Alternative: move everything into VMs on two large servers

Virtualisation can consolidate servers, improve hardware utilisation, simplify snapshots and speed up deployment. However, hypervisor scheduling, virtual I/O and resource contention add overhead and variability. A monitor alarm path requiring a guaranteed 10 ms deadline should not depend on a general-purpose VM on a shared host unless the hypervisor, CPU pinning, device passthrough, real-time scheduling and worst-case latency have been rigorously validated. A host failure could affect many VMs at once, creating a large failure domain. Two hosts can provide fail-over only if storage, networking, quorum and application recovery are also designed correctly.

Memory overhead includes hypervisor memory, guest OS memory and per-VM overhead; processor overhead includes virtual CPU scheduling and emulated I/O. Consolidation can reduce cost but increases common-mode risk. Recommended decision: virtualise ordinary application, test, reporting and some server workloads where isolation and performance are acceptable; keep safety-critical monitor controllers on dedicated validated hardware and retain independent redundant infrastructure for the database. Complexity of managing VMs grows with the

number of guests and their resource dependencies; the main trade-off is utilisation versus isolation and predictable latency.

QUESTION FROM THE PROVIDED ASSIGNMENT

Environment | Type / structure | Services, protection and I/O

Bedside monitors | Hard RTOS; microkernel preferred | Priority scheduling, bounded interrupt latency, sensor/alarm drivers, watchdog, secure boot, least privilege, local alarms; interrupt-driven sensor input and local buffering.

Nurses’ tablets | Managed handheld OS; Android Enterprise or iPadOS | Touch UI, power management, encryption, MDM, app sandbox, MFA, role-based access; Wi-Fi with secure API access and controlled local cache.

Doctors’ desktops | Managed desktop OS; Windows or Linux hybrid/modular architecture | Clinical GUI, application execution, endpoint protection, MFA, audit, patch management; network and local display I/O.

Patient-record database | Linux server OS with modular kernel; clustered active-passive or appropriately designed HA cluster | Database services, journaling, replication, backups, monitoring, RBAC, encryption at rest/in transit, audit; high-speed SSD/NVMe and redundant network.

MRI server | Linux multiprocessor SMP server | Parallel processing, large RAM, fast storage, job scheduling, access control; DMA for high-volume scanner transfers.

Partner hospitals | Distributed services over secure network | Authenticated APIs, encryption, mutual trust/identity federation, audit, replication and retry; local operation continues during network outages.